The documentation and process conversion measures necessary to comply with this revision shall be completed by 27 February 2001

INCH-POUND

MIL-PRF-19500/343E
28 November 2000
SUPERSEDING
MIL-PRF-19500/343D
10 December 1997

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON,
LOW POWER, TYPES 2N2857 AND 2N2857UB
JAN, JANTX, JANTXV, AND JANS, JANHC and JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, low power, ultra-high frequency transistors. Four levels of product assurance are provided for each device type and two levels for unencapsulated chips as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figures 1 (TO-72) and 2 (surface mount) and figure 3 (JANHC, JANKC).

1.3 Maximum ratings.

P_T (1) T_A = +25°C	P_T (2) T_C = +25°C	I_C	V_{CBO}	V_{CEO}	V_{EBO}	T_J and T_{STG}
mW	MW	mA dc	V dc	V dc	V dc	°C
200	300	40	30	15	3	-65 to +200

(1) Derate linearly 1.14 mW/°C for TA > +25°C.
(2) Derate linearly 1.71 mW/°C for TC > +25°C.

1.4 Primary electrical characteristics at T_A = +25°C.

	h_{FE1}	$\lvert h_{fe} \rvert$	C_{cb}	F	G_{pe}	$r_b' C_c$
	V_{CE} = 1 V dc I_C = 3 mA dc	V_{CE} = 6 V dc I_C = 5 mA dc f = 100 MHz	V_{CB} = 10 V dc I_E = 0 f = 100 kHz ≤ f≤ 1 MHz	V_{CE} = 6 V dc I_C = 1.5 mA dc f = 450 MHz R_g = 50 Ω	V_{CE} = 6 V dc I_C = 1.5 mA dc f = 450 MHz	V_{CB} = 6 V dc I_E = 2 mA dc f = 31.9 MHz
			pf	db	db	ps
Min	30	10	---	---	12.5	4
Max	150	21	1.0	4.5	21	15

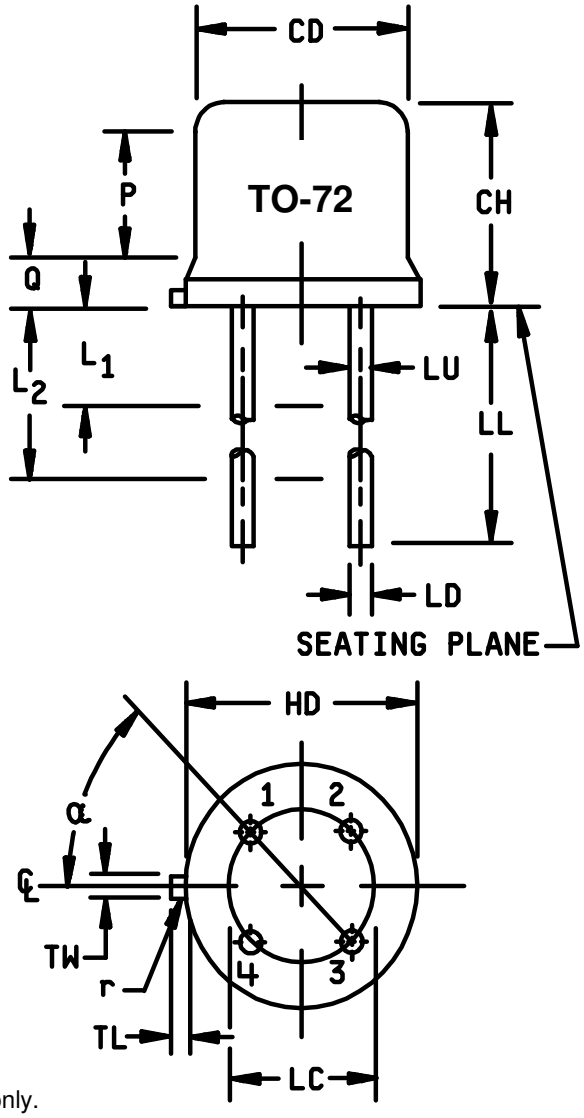
Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	5
CH	.170	.210	4.32	5.33	
HD	.209	.230	5.31	5.84	5
LC	.100 TP		2.54 TP		7,8
LD	.016	.021	.406	.533	7,8
LL	.500	.750	12.70	19.05	7,8
LU	.016	.019	.41	.48	
L1		.050		1 .27	
L2	.250		6.35		
P	.100		2.54		
Q		.040		1.02	5
TL	.028	.048	.71	1.22	
TW	.036	.046	.91	1.17	
r		.007		.18	
α	45° TP				

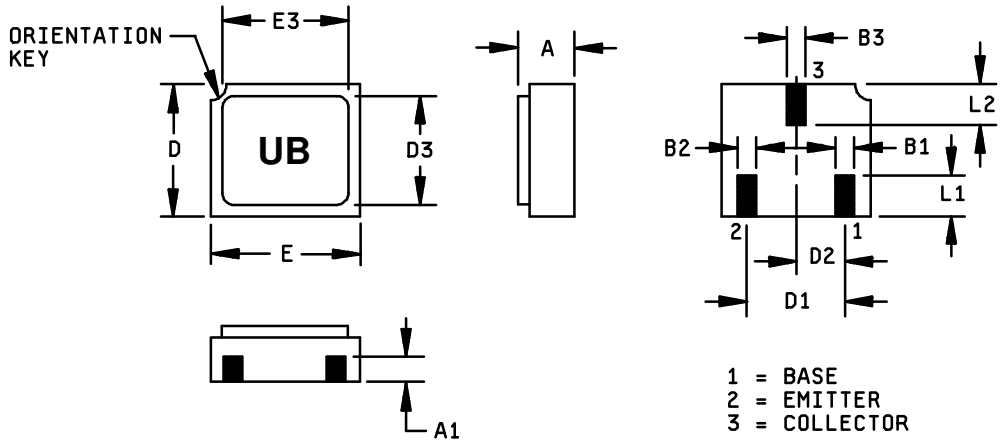


NOTES:

1. Dimension are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, TH shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gauge and gauging procedure shown in figure 2.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All four leads.
9. Dimension r (radius) applies to both inside corners of tab.
10. In accordance with ANSI Y14.5M, diameters are equivalent to Φx symbology.
11. Lead 1 = emitter, lead 2 = base, lead 3 = collector, lead 4 = case (electrically connected).

FIGURE 1. Physical dimensions for 2N2857, (TO-72).

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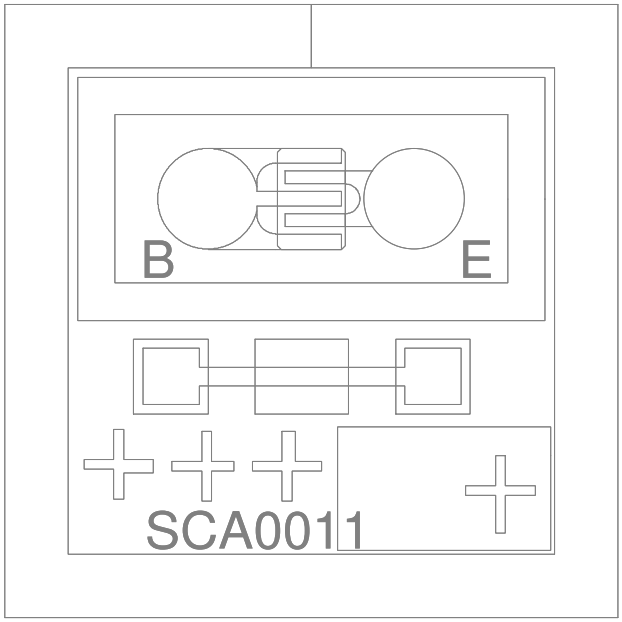


Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.046	.056	0.97	1.42	
A1	.017	.035	0.43	0.89	
B1	.016	.024	0.41	0.61	
B2	.016	.024	0.41	0.61	
B3	.016	.024	0.41	0.61	
D	.085	.108	2.41	2.74	
D1	.071	.079	1.81	2.01	
D2	.035	.039	0.89	0.99	
D3	.085	.108	2.41	2.74	
E	.115	.128	2.82	3.25	
E3	---	.128	---	3.25	
L1	.022	.038	0.56	0.96	
L2	.022	.038	0.56	0.96	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Physical dimensions, surface mount (UB version).

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Die size: ----------0.016 x 0.016 inches
Die thickness:---0.008 ± 0.0016 inches
Base pad:--------0.0023 x 0.0023 inches
Emitter pad:-----0.0023 x 0.0023 inches
Back metal:-----Gold, 6500 ± 1950 Ang
Top metal:-------Aluminum, 17500 ± 2500 Ang
Back side:-------Collector
Glassivation:----SiO_2, 7500 ± 1500 Ang

FIGURE 3. JANHC and JANKC (A-version) die dimensions

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

DEPARTMENT OF DEFENSE

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Services (DAPS), Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-19500.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.4).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1, 2 and 3.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups of table I herein.

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3.7 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, marking on the UB package may be omitted from the body, but shall be retained on the initial container.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3c	Thermal impedance (see 4.3.2)	Thermal impedance (see 4.3.2)
7	(1)	(1)
9	"I_{CES}, h_{FE1}".	Not applicable
10	48 hours minimum	48 hours minimum
11	I_{CES}, h_{FE1}; ΔI_{CES} = 100 percent of initial value or 5 nA, whichever is greater; Δh_{FE1} = 15 percent	I_{CES}, h_{FE1}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CES} = 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE1} = 15 percent	Subgroup 2 of table I herein; ΔI_{CES} = 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE1} = 15 percent

(1) Hermetic seal test shall be performed in either screen 7 or screen 14.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

V_{CB} = 10 Vdc, P_T = 200 mW at, T_A = Room ambient as defined in the general requirements of MIL-STD-750, 4.5.

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.3.2. Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500, Discrete Semiconductor Die/Chip Lot Acceptance. Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.

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4.3.3 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with method 3131, MIL-STD-750.

a. I_M measurement current..... 5 mA.

b. I_H forward heating current ..... 50 mA (min).

c. t_H heating time..... 20 - 30 ms.

d. t_{md} measurement delay time ..... 60 µs max.

e. V_{CE} collector-emitter voltage ..... 10 V dc minimum.

The maximum limit for $Z_{\theta JX}$ under these test conditions are $Z_{\theta JX}$ (max) = 75 °C/W.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed in accordance with E.5.3.1d of MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table VIa (JANS) of MIL-PRF-19500 and 4.4.2.1 herein. See 4.4.2.2 herein for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) and delta requirements shall be in accordance with the steps of table II herein and as specified in the notes for table II.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Conditions
B4	1037	V_{CB} = 10 V dc
B5	1027	$P_D \geq 75$ percent of maximum rated P_T (see 1.3). Option 1 - 96 hrs min, sample size in accordance with table VIa MIL-PRF-19500, adjust T_A to achieve T_j = +275°C minimum. Option 2 - 216 hrs min., sample size = 45, c = 0; adjust T_A to achieve T_j = +225°C minimum. (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.)

4.4.2.2 Group B inspection (JAN, JANTX and JANTXV). Separate samples may be used for each step. In the event of a group B failure, the manufacturer may pull a new sample at double size from either the failed assembly lot or from another assembly lot from the same wafer lot. If the new “assembly lot” option is exercised, the failed assembly lot shall be scrapped.

Step	Method	Conditions
1	1027	Steady state life: Test condition B, 340 hours, V_{CB} = 10 V dc; maximum rated power shall be applied and ambient temperature adjusted to achieve T_J = 150°C minimum; n = 45 devices, c = 0. For small lots, n = 12 devices, c = 0

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2	1027	The steady state life test of step 1 shall be extended to 1,000 hours for each die design. Samples shall be selected from a wafer lot every twelve months of wafer production. Group B, step 2 shall not be required more than once for any single wafer lot. n = 45, c = 0.
3	1032	High temperature life (non-operating), t = 340 hours, T_A = +200°C. n = 22, c = 0.

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Must be chosen from an inspection lot that has been submitted to and passed group A conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection, JANS. Group C inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table VII of MIL-PRF-19500, and in 4.4.3.1 herein (JANS). See 4.4.3.2 herein for JAN, JANTX, and JANTXV group C testing. Electrical measurements (end points) and delta requirements shall be in accordance with the steps of table II herein as specified in the notes for table II.

4.4.3.1 Group C inspection, table VII (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E; not applicable for UB devices.
C6	1026	V_{CB} = 10 V dc, 1,000 hours; maximum rated power shall be applied and ambient temperature adjusted to achieve T_J = +150°C minimum. N = 45 devices, c = 0. For small lots, n = 12 devices, c = 0."

4.4.3.2 Group C inspection, JAN, JANTX, and JANTXV, table VII of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E; not applicable for UB devices.
C6		Not applicable

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be pulled prior to the application of final lead finish. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Noise figure. The noise figure shall be measured using commercially available test equipment and its associated standard test procedures (see figure 4).

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical 3/ examination	2071	n = 45 devices, c = 0				
Solderability 3/, 4/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/, 4/, 5/	1022	n = 15 devices, c = 0				
Temp cycling 3/, 4/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Heremetic seal 4/ Fine leak Gross leak	1071	n = 22 devices, c = 0				
Electrical measurements 4/		Group A, subgroup 2				
Bond strength 3/, 4/	2037	Precondition T_A = +250°C at t = 24 hrs or T_A = +300°C at t = 2 hrs n = 11 wires, c = 0				
Subgroup 2						
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 3 mA dc	$V_{(BR)CEO}$		15	V dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 15 V dc	I_{CBO1}		10	nA dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 30 V dc	I_{CBO3}		1	µA dc
Emitter to base cutoff current	3061	Bias condition D V_{EB} = 3 V dc	I_{EBO1}		10	µA dc
Forward-current transfer ratio	3076	V_{CE} = 1 V dc; I_C = 3 mA dc	h_{FE1}	30	150	
Collector-emitter saturation voltage	3071	I_C = 10 mA dc; I_B = 1 mA dc;	$V_{CE(sat)}$		0.4	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 10 mA dc; I_B = 1 mAdc;	$V_{BE(sat)}$		1.0	V dc

See footnotes at end of table.

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TABLE I. Group A inspection, - continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 (continued)						
Collector to emitter cutoff current	3041	Bias condition C; V_{CE} = 16 V dc	I_{CES}		100	nA dc
Subgroup 3						
High temperature operation		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 15 V dc	I_{CBO2}		1.0	µA dc
Low temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 1 V dc; I_C = 3 mA dc	h_{FE2}	10		
Subgroup 4						
Noise figure		V_{CE} = 6 V dc; I_C = 1.5 mA dc; f = 450 MHz; R_g = 50 Ω; (case lead grounded); (see 4.5.2 and figure 4)	F		4.5	dB
Magnitude of common-emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 6 V dc; I_C = 5 mA dc; f = 100 MHz; (case lead grounded)	h_{fe}	10	21	
Small-signal short-circuit forward current transfer ratio	3206	V_{CE} = 6 V dc; I_C = 2 mA dc; (case lead floating)	h_{fe}	50	220	

See footnotes at end of table.

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TABLE I. Group A inspection, - continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 (continued)						
Collector to base - feedback capacitance		V_{CB} = 10 V dc; I_E = 0 mA dc; 100 kHz ≤ f ≤ 1 MHz (see 4.5.1)	C_{cb}		1.0	pF
Small-signal power gain	3256	V_{CE} = 6 V dc; I_E = 1.5 mA dc; f = 450 MHz	G_{pe}	12.5	21	dB
Collector - base time constant		V_{CB} = 6 V dc; I_E = 2 mA dc; f = 31.9 MHz (see figure 5)	$r_b' C_c$	4	15	pF

1/ For sampling plan (unless otherwise specified), see MIL-PRF-19500.
2/ For resubmission of failed subgroup A1, double the sample size of the failed test or sequence of tests. A failure in group A, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.

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TABLE II. Groups A, B, and C electrical end-point inspection measurements. 1/ 2/ 3/ 4/

Step	Inspection	MIL-STD-750		Symbol	Limit		Unit
		Method	Conditions		Min	Max	
1.	Collector-base cutoff current	3036	Bias condition D; V_{CB} = 15 V dc	I_{CBO1}		10	nA dc
2.	Collector - base cutoff current	3036	Bias condition D; V_{CB} = 15 V dc	I_{CBO2}		20	nA dc
3.	Collector to emitter voltage (saturated)	3071	I_C = 10 mA dc; I_B = 1.0 mA dc pulsed	$V_{CE(sat)}$		0.4	V dc
4.	Base-emitter voltage (saturated)	3066	Test condition A I_C = 10 mA dc; I_B = 1 mA dc	$V_{BE(sat)}$		1.0	V dc
5.	Forward current transfer ratio	3076	V_{CE} = 1 V dc; I_C = 3 mA dc	h_{FE1}	30	150	
6.	Collector-base cutoff current	3036	Bias condition D V_{CB} = 15 V dc	ΔI_{CB01}	100 percent of initial value or 5 nA dc, whichever is greater.		
7.	Forward current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 3 mA dc;	Δh_{FE1}	± 25percent change from initial value.		
8.	Collector to emitter voltage(saturated)	3071	I_C = 10 mA dc; I_B = 1.0 mA dc	$\Delta V_{CE(sat)}$	± 50 mV dc change from initial value.		

1/ The electrical measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:
   a. Subgroup 3, see table II herein, steps 1, 3, 4, 5, and 6.
   b. Subgroups 4 and 5, see table II herein, steps 2, 3, 4, 5, and 8.

2/ The electrical measurements for group B, 4.4.2.2 herein (JAN, JANTX, and JANTXV) are as follows:
   Steps 1, 2, and 3 of 4.4.2.2, see table II herein, steps 2, 6, 7, and 8.

3/ The electrical measurements for table VII (JANS) of MIL-PRF-19500 are as follows:
   a. Subgroup 2 and 3, see table II herein, steps 1, 3, 4, 5, and 6 (for JANS).
   b. Subgroup 6, see table II herein, steps 2, 3, 4, 5, 6, 7, and 8 (for JANS).

4/ The electrical measurements for group C, 4.4.3.2 herein (JAN, JANTX, and JANTXV) are steps 1, 3, 4 and 5.

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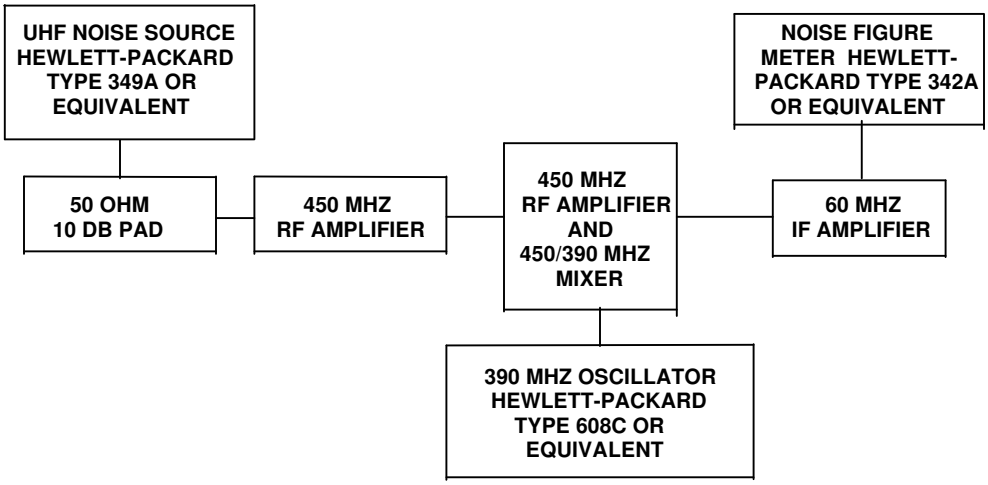
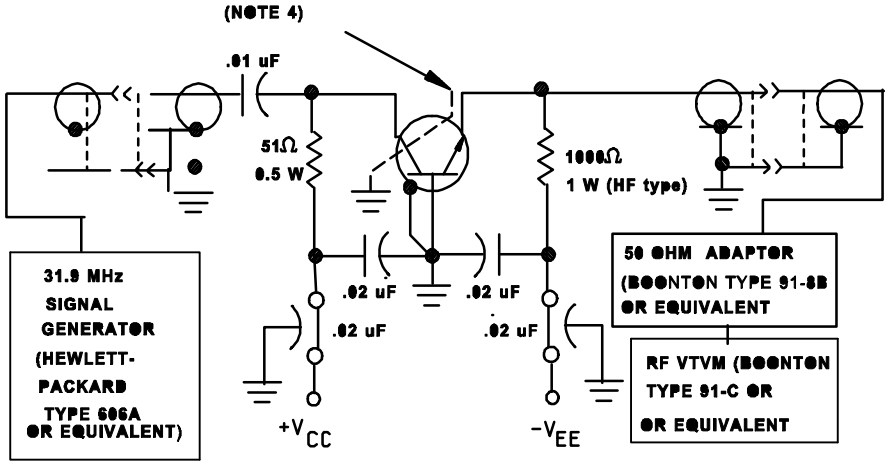


FIGURE 4. Block diagram for noise figure test.



Procedure:

1. With a short circuit applied between the collector and emitter terminals adjust 31.9 MHz input for 0.5 V RMS at emitter terminal of transistor.
2. After removing the short circuit between the collector and emitter circuit, insert unit to be tested and adjust V_{CC} and V_{EE} for V_{CB} = 6 V dc, I_E = 2 mA dc.
3. Read $r_b'C_C$ on RF voltmeter scale ($r_b'C_C$ in picoseconds = 10 times meter indication in millivolts) (1 millivolt = 10 picoseconds).
4. External interlead shield to isolate the collector lead from the emitter and base lead.

FIGURE 5. Collector-base time constant test circuit (an equivalent circuit may be used).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).

c. Lead formation and finish may be specified (see 3.4.1).

d. Type designation and product assurance level.

e. For die acquisition, the JANHC or JANKC letter version shall be specified (see figure 5).

f. Surface mount designation if applicable.

g. Packaging requirements (see 5.1).

6.3 Supersession information. Devices covered by this specification supersede the manufacturers' and users' Part or Identifying Number (PIN). The term PIN is equivalent to the term part number which was previously used in this specification. This information in no way implies that manufacturers' PINs are suitable as a substitute for the military PIN.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, P.O. Box 3990, Columbus, OH 43216-5000

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6.5 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N2857) will be identified on the QPL.

Die ordering information	
PIN	Manufacturer
	34156
2N2857	JANHCA2N2857, JANKCA2N2857

6.6. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 13, 19, 99

Preparing activity:
DLA - CC

(Project 5961-2407)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/343E	2. DOCUMENT DATE 28 November 2000

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW POWER, TYPES 2N2857 AND 2N2857UB JAN, JANTX, JANTXV, AND JANS, JANHC and JANKC

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (PLEASE PRINT)	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE Commercial DSN FAX EMAIL 614-692-0510 850-0510 614-692-6939 alan_barone@dscc.dla.mil
c. ADDRESS *(Include Zip Code)* Defense Supply Center Columbus, DSCC-VAC P.O. Box 3990 Columbus, Ohio 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703)767-6888 DSN 427-6888

DD Form 1426, FEB 1999 (EG) PREVIOUS EDITIONS ARE OBSOLETE. WHS/DIOR, Feb 99

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